

Rock Evidence Lab: Look Before You Label

BUILD • Science • 40 minutes

I can compare rock specimens using observed properties and create a group using a stated rule.

Name: _____ Date: _____

You may write, type, say, sign, point or direct a partner. An adult can read the words and record your exact response. Keep the choice and explanation your own.

Words for this lesson

Word	Meaning
property	a feature we can notice or test
observe	look closely and record
classify	put into groups with a rule

First idea

Two rocks are both grey. Is that enough evidence to place them in the same useful group?

A. Yes. One shared colour is all we need for every purpose.

B. Not yet. Observe the same extra features on both rocks, then state a grouping rule.

C. Guess both names first and make the evidence fit later.

My first idea and reason:



Coded rock A: speckled with crystals

We do • choose, explain, check

Which line is an observation that can be checked on the supplied specimen?

A. Several large crystals are visible in the speckled surface.

B. It must be granite because that is my first guess.

C. It will make the best roof because it looks strong.

My choice: _____ Evidence or reason:

Coded Rock Feature Lab

Place each coded observation under its strongest named visible feature. The name is not the target.

Read or hear one card. Choose a group. Give the evidence. A partner checks your reason before you move on. Point or write card numbers; cutting is optional.

Group	Card numbers and evidence
VISIBLE CRYSTALS	
VISIBLE GRAINS	
BANDS / FLAT PLANES	
LARGE HOLES	
FINE • NO LARGE FEATURE	

Activity cards

1 • Code A	2 • Code B
Speckled surface with several visible interlocking crystals.	Dark, fine-looking surface with no large visible grains, bands or holes.

3 • Code C

Rough tan surface with many visible sand-sized grains and faint bands.

4 • Code D

Pale, mostly fine-looking surface with no large crystals, bands or holes in this view.

5 • Code E

Blue-grey fine-looking specimen with clear flat, sheet-like planes.

6 • Code F

Pale rough-looking surface with many clearly visible large holes.

7 • Mystery image T

Brown-grey surface with visible bands, no large crystals and no large holes.

Choose one visible feature as a rule. Which codes belong in the yes group, and which evidence supports each placement?

Your lesson record

Inspect or direct the inspection of coded rock specimens, compare two using the same features and create two groups with a stated rule.

Model helps us see

Coded macro views slow down naming and make the observation-to-rule sequence visible.

Model does not show

Appearance alone cannot prove an unknown rock's identity or hidden properties. Lighting, scale, weathering and natural variation can change how a specimen looks.

Supported

Compare two coded macro cards with a four-word feature bank, then place them using one shown rule.

Use: I can see ___. Both have ___ / only ___ has ___. Adult exact scribing is valid.

Standard

Observe four to six coded specimens, compare a pair and create two groups using one stated property rule.

Record the rule before moving specimens. Use only evidence that another person can check.

Stretch

Create two different valid groupings, explain why both work and state why neither proves every specimen's identity.

Separate classification under a rule from identification of an unknown rock.

Evidence target

A Rock Comparison Ticket with two codes, two observed properties, one similarity, one difference and one explicit grouping rule.

Exit, Audience and evidence • W9A

Supported: Point and complete: I observed ___. My rule was ___.

Standard: Compare two coded specimens and defend one group using observed properties.

Stretch: Explain how a new valid rule could regroup the same specimens and why appearance cannot prove every name.

What I said, and what it changed

Next I will _____.

Adult witness note

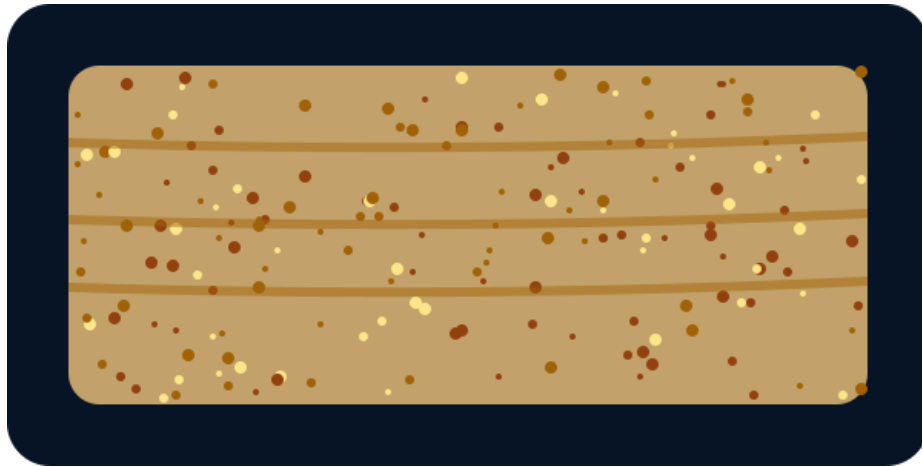
What was observed, and with what level of independence?

My evidence and explanation

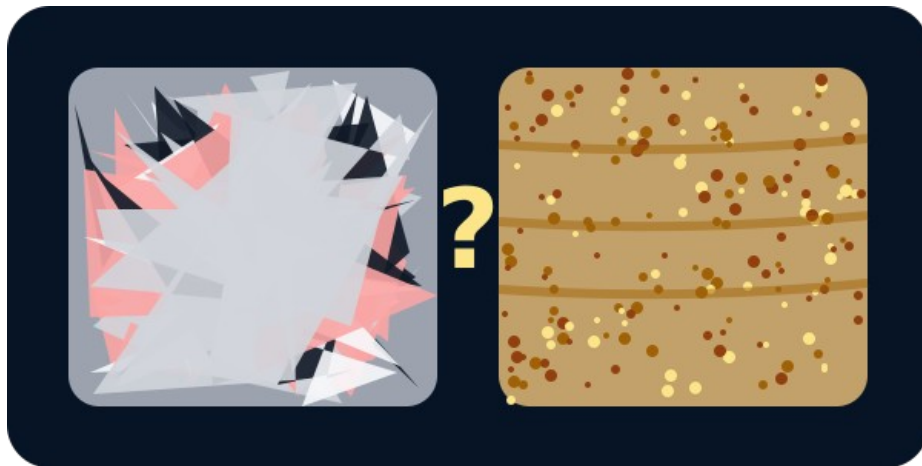
What does this evidence not show?

Visual reference cards

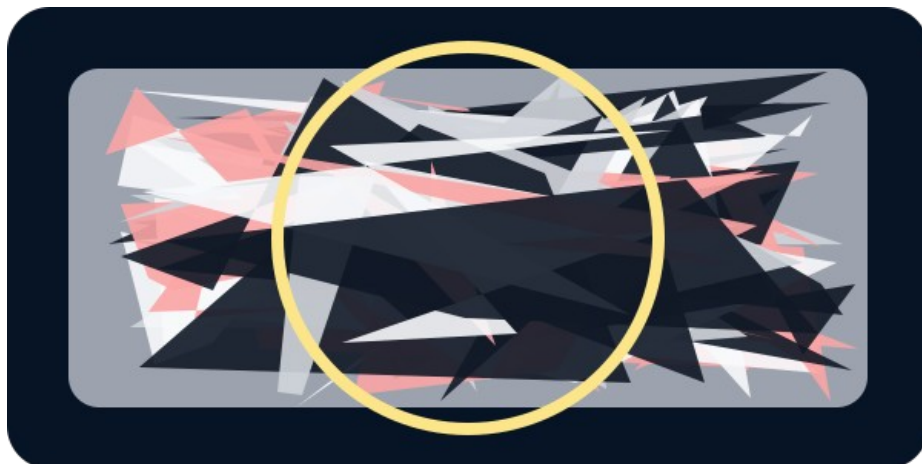
These are diagrams or model views from the uploaded lesson. Use their labels and the card descriptions; they are not photographs of your own results.



Coded rock C: sandy grains with faint bands



Rock A beside rock C with a question mark



Magnified view of the chosen coded rock

Exit • one next step

After the adult has listened, what will you keep, improve or check next?
